

Appendix D — Inference Loop Mathematics

This appendix formalizes the discrete-time active inference loop executed by `cogant` process on the extracted A/B/C/D matrices and reported in Section 5 (Table of VFE traces) and `../cogant/docs/evaluation/EMPIRICAL_CLAIM.md`. The formalism follows Da Costa et al. (2020) and the `pymdp` reference (Heins et al., 2022); we restate it here in notation consistent with COGANT's `cogant.process` module.

D.1 POMDP formulation

The extracted model is a discrete-time Partially Observable Markov Decision Process (S , O , A , γ , A_mat , B_mat , C_mat , D_mat) with:

$S = \{s_1, \dots, s_{|S|}\}$ — finite set of hidden states. Cardinality is the product of factor cardinalities: $|S| = \prod_f |S_f|$.

$O = \{o_1, \dots, o_{|O|}\}$ — finite set of observations. For multi-modality models, $|O| = \prod_m |O_m|$.

$A = \{1, \dots, |A|\}$ — finite set of discrete actions (control states).

